

Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection

Trong Nhan Nguyen[†]

Industry 4.0 Convergence

Bionics Engineering

Department Biomedical Engineering

Pukyong National University

Busan 48513, Republic of Korea

nguyennhan8521@gmail.com

Vu Hoang Minh Doan[†]

Smart Gym-Based

Translational Research Center

for Active Senior's Healthcare

Pukyong National University

Busan 48513, Republic of Korea

doanvuhoangminh@gmail.com

Tan Hung Vo

Smart Gym-Based

Translational Research Center

for Active Senior's Healthcare

Pukyong National University

Busan 48513, Republic of Korea

tanhung0506@gmail.com

Jaeyeop Choi

Smart Gym-Based

Translational Research Center

for Active Senior's Healthcare

Pukyong National University

Busan 48513, Republic of Korea

jaeyeopchoi@pknu.ac.kr

Junghwan Oh

Senior Member, IEEE

Ohlabs Corp, Industry 4.0 Convergence

Bionics Engineering

Department of Biomedical Engineering

Pukyong National University

Busan 48513, Republic of Korea

jungoh@pknu.ac.kr

Abstract—Despite significant advancements in super-resolution (SR) techniques, their application in industrial nondestructive testing (NDT) remains challenging due to stringent requirements for accuracy and reliability. Among NDT technology, scanning acoustic microscopy (SAM) is widely used in semiconductor industries to generate high-resolution images for detecting defects in increasingly miniaturized and complex components. The integration of SR into SAM has the potential to enhance image quality without increasing scanning time, making it a promising solution for industrial applications. However, a main problem in achieving optimal SR performance in this field is the limited availability of training data, as scanning images of the electronic samples are often confidential. We introduce a method for generating synthetic low-resolution datasets from high-resolution images using a high-order degradation model to mitigate this data scarcity problem. To address the demand for fast, high-resolution imaging in SAM, we present an SR approach that combines Generative Adversarial Networks (GANs) with Transformer-based architectures, leveraging Transformers as the generator network to enhance the GAN model's performance. Our model, named SAMTransGAN, incorporates transfer learning and a dynamic loss function that balances pixel-wise, perceptual, and adversarial loss for optimal results. Our SAMTransGAN achieves better performance, attaining Peak Signal-to-Noise Ratio (PSNR) of 28.158 dB and a Structural Similarity Index Measure (SSIM) of 0.835 on the testing dataset, surpassing all other compared models. We also present a possible deployment framework for our proposed SAMTransGAN on the web page as an efficient and compact SR solution for the NDT industry. Our model increases image resolution while preserving critical feature details, demonstrating its practical applicability in industrial settings.

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[†] These two authors contribute equally to the work.

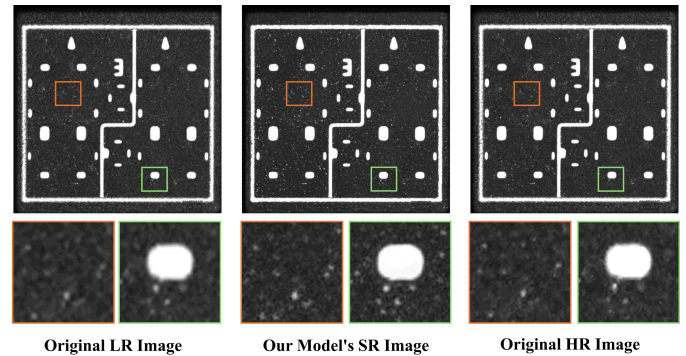


Fig. 1. Qualitative comparisons between the low-resolution (LR) image, the super-resolution (SR) image generated by our proposed SAMTransGAN, and the high-resolution (HR) image.

Index Terms—Scanning Acoustic Microscopy; Super Resolution; Transformer.

I. INTRODUCTION

WITH the rapid evolution of the semiconductor industry, electronic devices have become increasingly innovative and complex. This sophistication in internal structures necessitates high-quality control to ensure reliability and performance. Non-destructive testing (NDT) techniques are generally employed to inspect these internal structures without causing damage to the samples. There are currently several popular approaches in this field, such as X-ray microscopy (XRM), Scanning acoustic microscopy (SAM), and Terahertz radiation [1].

Among them, SAM is well-known as a reliable, fast, large-area, and user-friendly inspection technique [2]. However, SAM's primary limitation is its relatively low image resolution compared to other methods. Achieving high-resolution (HR) SAM images often requires compromises, such as reducing penetration depth or increasing scanning time. Typically, higher resolution is obtained by employing high-frequency transducers, which inherently have shallow penetration due to the physical property of high-frequency ultrasound waves. This limitation restricts SAM's ability to effectively inspect multi-layered or thick samples, which is commonly encountered in industrial electronics fields. To overcome these limitations and enhance resolution without sacrificing other critical scanning characteristics, advanced techniques like super-resolution (SR) offer a promising solution.

The SR is an advanced image processing technique widely used in various fields, but its applications in the industrial NDT ultrasound field are still limited [3, 4]. One study utilized the SAM system to inspect the surface of coins, enhancing the resolution of scanning images using a transformer-based architecture [5]. This research demonstrated the potential of combining high-frequency transducers with transfer learning. However, the study's applicability to industrial electronic inspections is limited due to shallow penetration and a lack of reliability testing. Another study employed a semi-supervised generative model for ultrasound image SR, but the training data were generated via numerical simulations, and the model was validated only on artificial samples [6]. While both approaches improved image resolution, their reliance on simulated data and artificial validation raises concerns about their practicality for real-world industrial applications.

Deep learning models for SR tasks commonly employ architectures such as Generative adversarial networks (GANs) and Transformers [7]. GANs excel in generating realistic textures and fine details through adversarial training, making them highly effective for visually appealing SR outcomes [8]. However, they are prone to artifacts and training instability due to the adversarial framework. Transformers, on the other hand, leverage self-attention mechanisms to capture long-range dependencies and global context, ensuring structural coherence and scalability [9]. Nevertheless, their application is computationally intensive and typically requires large datasets, making them less efficient than GANs in certain scenarios. In this study, we propose an approach that integrates a transformer-based generator with a GAN architecture to advance SR imaging for SAM. This hybrid model combines the global dependency modeling capabilities of Transformers with the fine-detail generation strength of GANs, addressing the limitations of traditional SAM methods. Our solution significantly enhances image quality by improving both structural accuracy and perceptual fidelity, accelerating the resolution of SAM images to meet industrial standards for electronic component inspection (Fig. 1).

The paper is organized as follows: Section 2 details our methodology, including the degradation network, generator model, and discriminator model on network architecture. Sec-

tion 3 presents the experiment setup, results, and deployment on the web page. Finally, Section 4 concludes the study and outlines future directions.

II. METHODOLOGY

A. Degradation Network

The SAMTransGAN model was trained using synthetic low-resolution (LR) images, generated from HR images through a degradation process [10]. As shown in Fig. 2, this process follows the classical degradation model, which incorporates three main components: blur, noise, and downsampling [11]. To introduce variability, the degradation pipeline utilizes a randomized sequence of operations across these components [12]. For HR images, the sequence $\{B_{\text{iso}}, B_{\text{aniso}}, D^s, N_G, N_{\text{JPEG}}, N_S\}$ is applied, where B_{iso} and B_{aniso} represent isotropic and anisotropic Gaussian blurring, respectively [13]. The downsampling operation, D^s , employs a scale factor s , chosen randomly from four methods: nearest-neighbor interpolation (D_{nearest}^s), bilinear interpolation (D_{bilinear}^s), bicubic interpolation (D_{bicubic}^s), and a hybrid down-up sampling approach ($D_{\text{down-up}}^s$). In the down-up approach, the image is first downsampled by s/a and then upsampled by a :

$$D_{\text{down-up}}^s = D_{\text{down}}^{s/a} D_{\text{up}}^a. \quad (1)$$

Finally, noise addition includes Gaussian noise (N_G), Joint Photographic Experts Group (JPEG) compression artifacts (N_{JPEG}), and simulated camera sensor noise (N_S) [14, 15].

B. Network Architecture

1) *Generator model*: The generator network architecture, as illustrated in Fig. 3, is composed of three main modules: shallow feature extraction, deep feature extraction, and image reconstruction. The shallow feature extraction module captures local low-level information using convolutional layers. The deep feature extraction module, based on the Transformer architecture, utilizes multi-head self-attention mechanisms (with N attention heads) to model long-range spatial dependencies more effectively than traditional Convolutional Neural Networks (CNNs). The image reconstruction module employs advanced sub-pixel upsampling techniques to enhance resolution efficiently. The integration of these modules allows the network to transform an LR input image into an HR output image by doubling the dimensions while preserving fine details and structural features for high-quality image reconstruction.

Shallow Feature Extraction: Starting with an LR image, shallow features [16] are extracted through a convolutional layer. This layer plays a critical role in initial visual processing, facilitating stable optimization and improving output quality. The primary aim of this module is to help the model capture low-frequency details, such as color and texture [17].

Deep Feature Extraction: The residual structure is utilized in this component, as it has been proven effective in single-image super-resolution (SISR) tasks, such as ESRGAN [17]. This module is designed to extract deep features through six Residual Shifted Window Transformer Blocks (RSTBs) [18], with each block followed by a convolutional layer [19]. Each

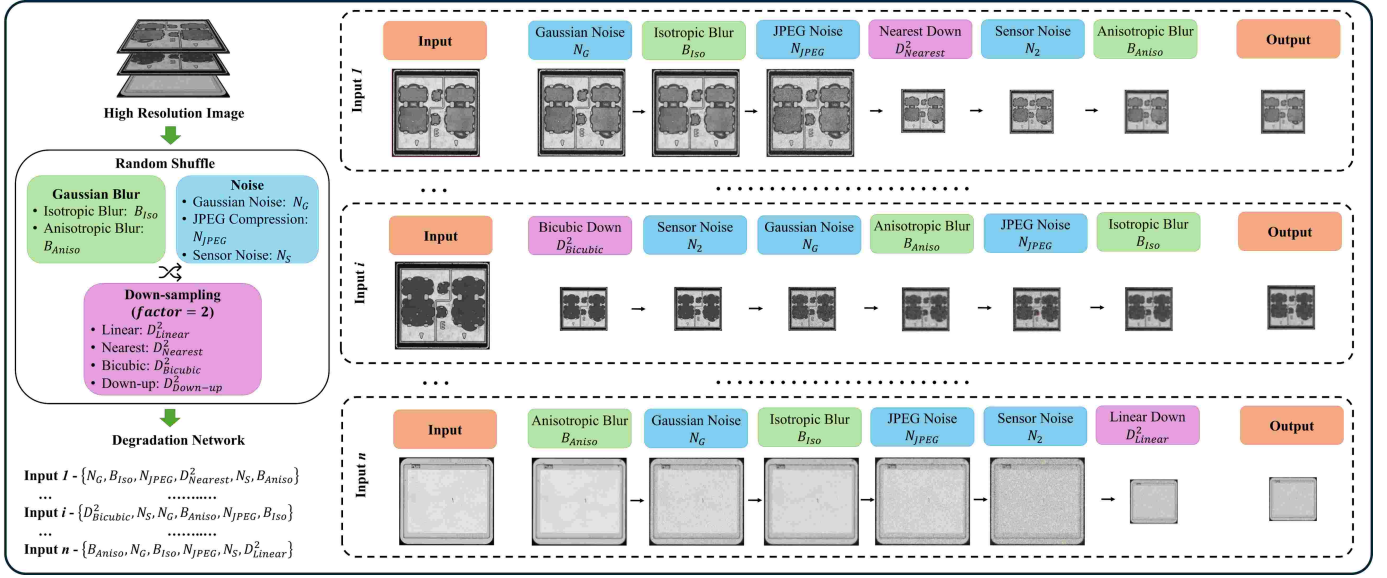


Fig. 2. Overview of the pure synthetic data generation process adopted in SAMTransGAN. For each HR image, the randomly shuffled degradation sequence $\{B_{Iso}, B_{Aniso}, D^s, N_G, N_{JPEG}, N_S\}$ is applied, followed by a downscaling operation with scale factor 2.

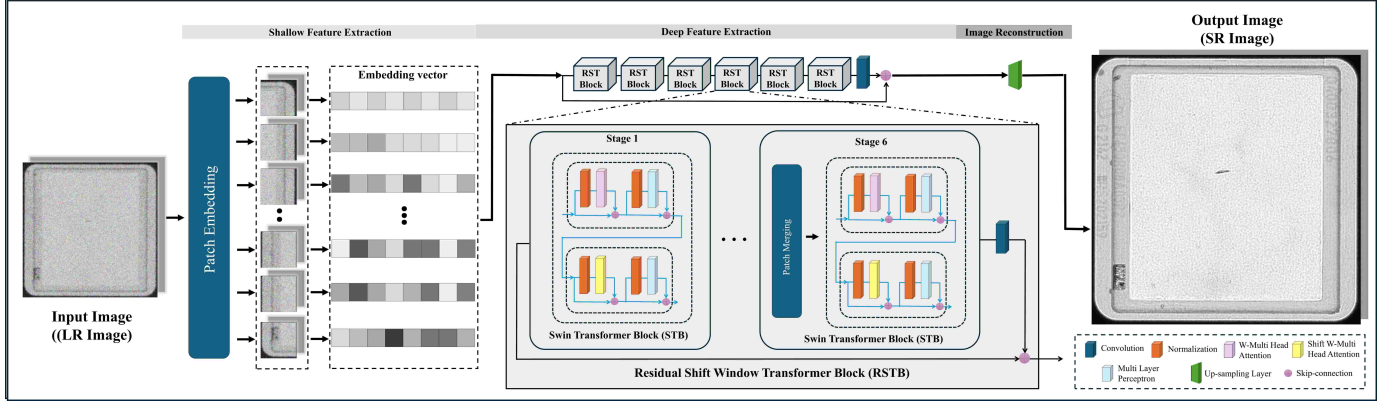


Fig. 3. The overall architecture of SAMTransGAN generator network and the structure of Residual Shift Window Transformer (RST). The figure illustrates a Transformer-based network for Single Image Super-Resolution (SISR), enhancing LR input images to produce HR output. It comprises three main stages: Shallow Feature Extraction, Deep Feature Extraction, and Image Reconstruction. The input image is processed through Patch Embedding, creating embedding vectors fed into the Residual Shift Window Transformer blocks (RSTBs), which incorporate Swin Transformer blocks (STBs) to model global and local dependencies. Features are progressively merged and refined across six stages, culminating in an up-sampling layer for final reconstruction.

RSTB comprises six Shifted Window Transformer Blocks (STBs) [20], and the final layer includes an additional convolutional layer to enhance feature representation. At the end of this process, a convolutional layer is included to further refine the combination of shallow and deep features.

Shifted Window Transformer Block (STB): The STB is based on the original Transformer layer's multi-head self-attention mechanism [21, 22], with key distinctions in local attention and the shifted window approach. Self-attention within each local window captures relationships between different patches of the image, computed as:

$$\text{Attention}(Q, K, V) = \text{SoftMax} \left(\frac{QK^T}{\sqrt{d}} + B \right) V \quad (2)$$

Here, Q , K , and V represent the query, key, and value matrices, respectively, where d is the dimension of the query/key.

The number of windows defined as $\frac{HW}{M^2}$, where H , W are the height and width of the input image, and M^2 refers to the number of patches in a window. $B \in \mathbb{R}^{M^2 \times M^2}$ is the learnable relative positional encoding [22]. The attention mechanism is computed concurrently across multiple heads, with the outputs from each head concatenated to construct the multi-head self-attention (MSA) framework. Following the attention computation, a Multi-Layer Perceptron (MLP) comprising two fully connected layers and a Gaussian Error Linear Unit (GELU) activation function [23] is employed to refine the features further. To enhance stability and efficiency, both the MSA and MLP modules are initially processed by Layer Normalization (LN) layers, and residual connections are integrated after each module to ensure effective information

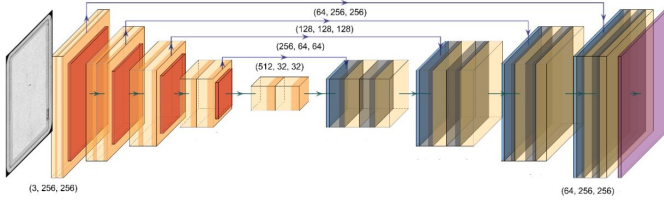


Fig. 4. The Discriminator Network used in the SAMTransGAN model.

flow and gradient propagation.

Image Reconstruction: The shallow features predominantly capture low-frequency information, while the deep features are tasked with recovering high-frequency details that may be lost. A long skip connection [16] facilitates the direct transmission of low-frequency information from the intermediate module to the reconstruction module [17]. Within the reconstruction module, a sub-pixel convolution layer [24] is employed to upsample the features, thereby enhancing the image resolution.

2) *Discriminator model:* The proposed discriminator leverages a U-Net-based architecture [25] enhanced with spectral normalization applied to all convolutional layers, as illustrated in Fig. 4. Spectral normalization constrains the Lipschitz constant, ensuring stable training and improved resilience to diverse degradation conditions [26].

By incorporating downsampling and upsampling paths with skip connections, the architecture provides detailed pixel-level feedback to the generator, aiding in the reconstruction of intricate image details. The addition of spectral normalization not only stabilizes training but also enhances the model's generalization, making it particularly effective for processing complex industrial SAM data [17, 27].

III. EXPERIMENTAL

A. Datasets and Implementation

1) *Datasets preparation:* All imaging data were acquired using the TSAM-400 SAM system (Ohlabs Corp, Busan, Korea) [28], a key NDT instrument comprising a scanner, pulser/receiver, digitizer, and workstation. The system enables X, Y, and Z-axis movement and supports transducers within a 5–200 MHz range; here, a 100 MHz transducer was employed. Data acquisition involved securing the transducer on the Z-axis, scanning along the Y-axis for B-scan cross-sections, and incrementally adjusting along the X-axis to create a comprehensive C-scan. This echo-mode raster scanning approach was applied to four industrial samples, generating paired datasets with 50 μm for HR and 100 μm for LR.

2) *Loss Function:* In this work, the SAMTransGAN model is trained by minimizing a total loss function, which is a weighted sum of three primary components: pixel-wise loss $\mathcal{L}_1$ [29], perceptual loss $\mathcal{L}_{\text{Percep}}$ [30], and adversarial loss $\mathcal{L}_{\text{Adver}}$ [17]. The total loss is expressed as:

$$L_{\text{total}} = \alpha\mathcal{L}_1 + \beta\mathcal{L}_{\text{Percep}} + \gamma\mathcal{L}_{\text{Adver}} \quad (3)$$

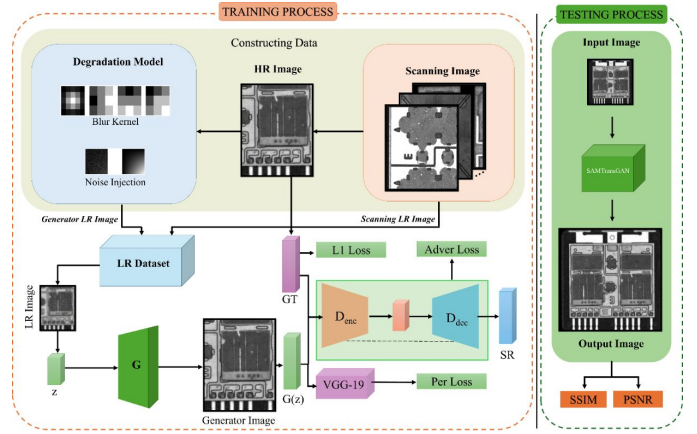


Fig. 5. Overview of the SAMTransGAN framework, including both the training and testing processes. During training, the degradation model generates synthetic LR images, which are combined with existing LR images to form the LR dataset for the generator model. The generator enhances the resolution of the input images, while the discriminator evaluates the outputs. The training process optimizes a total loss function, ultimately producing SR images. In the testing process, LR images at a step size of 100 μm are used as inputs to SAMTransGAN, which enhances their resolution. Image quality is evaluated using PSNR and SSIM metrics.

where α , β , and γ are hyperparameters controlling the relative proportion of each loss component. The pixel-wise loss $\mathcal{L}_1$ is essential for preserving fine-grained image details; the perceptual loss $\mathcal{L}_{\text{Percep}}$ ensures high-level semantic features are retained, and the adversarial loss $\mathcal{L}_{\text{Adver}}$ helps improve the visual authenticity of the generated images.

3) *Implementation details:* The training process is divided into three stages. In the first stage, the process begins with the utilization of a pre-trained weight model that has been trained on a substantial dataset (DIV2K dataset [31]), specifically designed for image SR tasks. In the second stage, the primary objective of this phase is to leverage transfer learning, allowing the model to capture complex and general features inherent in large datasets [16]. In the subsequent training phase as illustrated in Fig. 5, the transformer model is refined using a more focused dataset, maintaining the use of the L_1 loss function with an emphasis on achieving high PSNR (PSNR-oriented) [32]. The goal during this stage is to facilitate the reconstruction of HR images that closely align with the ground truth in terms of numerical accuracy, thereby minimizing discrepancies between the generated outputs and the original images. This phase also prioritizes the recovery of essential image features, such as edges and fine details, with a focus on precision and minimal noise introduction. Ensuring high structural accuracy sets a solid foundation for the next stage of development. In the final training phase, the GAN model is employed, incorporating a combination of loss functions, including L_1 loss, perceptual loss, and adversarial loss [33]. This phase aims to enhance the perceptual quality of the generated images, making them appear more natural and realistic while preserving the structural integrity established in the previous training phase.

Our SAMTransGAN model is trained utilizing a composite

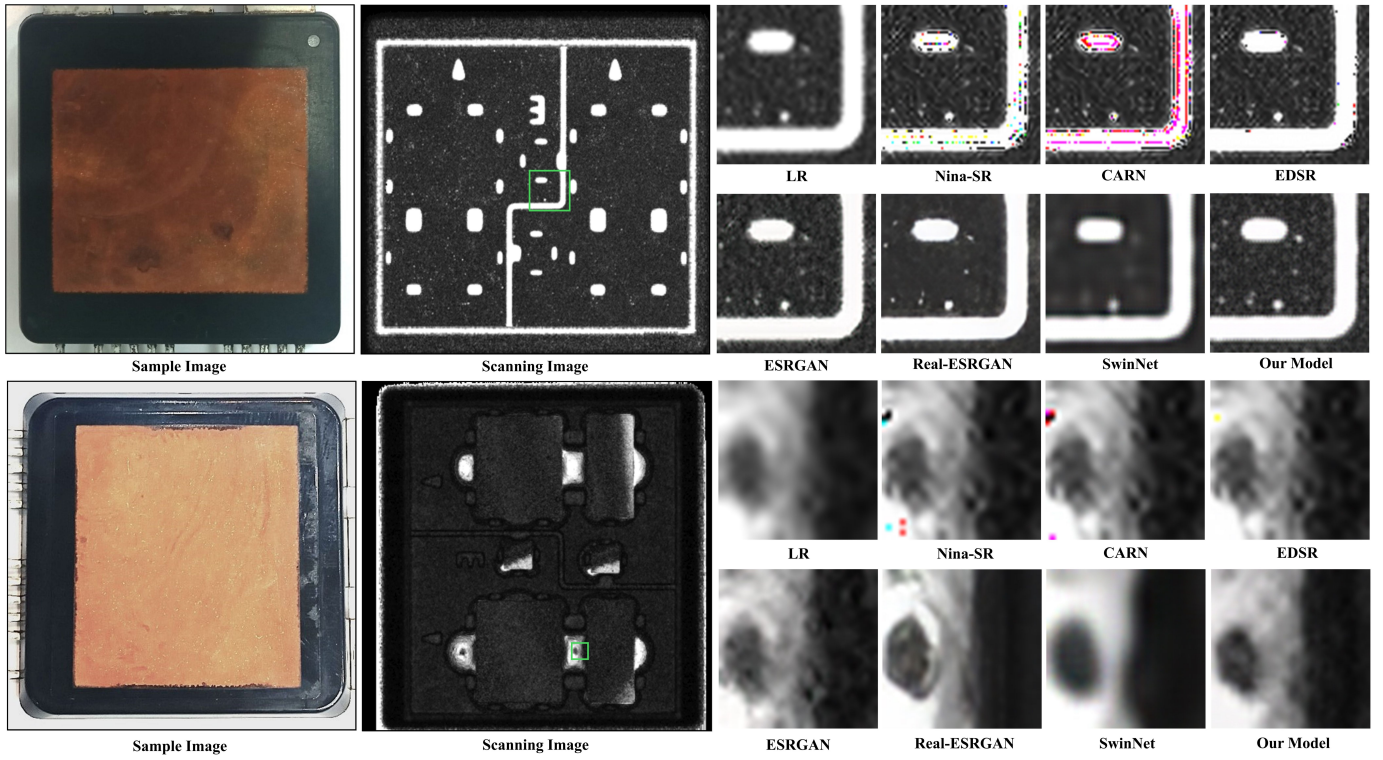


Fig. 6. Quantitative results of different methods on the test dataset with an up-sampling scale factor of 2. The images include a ceramic layer image and a solder layer image of a power MOSFET sample.

loss function comprising L_1 loss, perceptual loss, and adversarial loss, with a learning rate set at 0.0002. The weights assigned to each loss component were fine-tuned to values of 1.0, 0.5, and 0.05, respectively, to optimize the generation of super-resolved images. The training procedure involved a batch size of 32, with 16 workers used for data loading to enhance efficiency. The model's architecture utilized the SwinIR framework [34], with an upscale factor of 2 and a window size of 8, allowing for effective handling of image resolution enhancement tasks. The training was conducted on a high-performance distributed GPU system [35], featuring both NVIDIA GeForce RTX 3090 Ti and RTX 4090 Ti GPUs, providing robust computational power. The network configurations were implemented using Python 3.9 and executed with CUDA 12.5, while PyTorch 2.3.1 was employed as the back-end framework for model development and training [36]. Additionally, the training utilized an exponential moving average decay rate of 0.999 for the generator network to stabilize the model updates. This setup ensured the availability of adequate computational power and a suitable software environment to facilitate the experimental processes effectively.

In our study, we evaluated our model's performance using two widely adopted metrics in Single Image Super Resolution (SSIR) research, namely Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) [37]. It is important to note that PSNR and SSIM are computed in the luminance (Y) channel of the YCbCr color space. Together,

PSNR and SSIM offer a comprehensive evaluation, reflecting both fidelity and perceptual quality [38].

B. Comparison with state-of-the-arts

In Fig. 6, we present a detailed qualitative comparison of several test images using state-of-the-art (SOTA) SR models. To validate the performance of our model and others, we utilized scanning images from specific layers of power MOSFET, which present complex textures and high-contrast regions critical for evaluation. CNN-based SR methods [39], such as Nina-SR [40], CARN [41], and EDSR [16], exhibit limitations in areas with complex features, often resulting in color distortions and the loss of fine details due to their reliance on local feature extraction through sliding windows. These methods particularly struggle to maintain accuracy in high-contrast areas, such as the white-colored region in the upper part of the sample images, where insufficient feature information leads to artifacts and diminished clarity. Among GAN-based models, ESRGAN [17] shows impressive capability in enhancing resolution and preserving texture, especially in backgrounds. However, it introduces occasional inaccuracies in detailed representation. Real-ESRGAN [32] refines image sharpness and highlights many object details, but it tends to over-smooth certain background regions, resulting in an unnatural appearance. Transformer-based models, like SwinIR [34], efficiently capture long-range dependencies, showing superior restoration of global features; however, they sometimes compromise finer details. In contrast, our SAMTransGAN built

TABLE I
QUANTITATIVE COMPARISON OF PSNR AND SSIM METRICS WITH STATE-OF-THE-ART METHODS ON THE TEST DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Image no.	Nina-SR		CARN		EDSR		ESRGAN		Real-ESRGAN		SwinNet		Our Model	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
1	21.269	0.896	26.223	0.922	31.526	0.921	36.780	0.902	37.060	0.936	35.588	0.919	38.324	0.933
2	21.113	0.912	31.754	0.940	32.099	0.941	35.697	0.916	28.425	0.897	33.924	0.910	32.630	0.932
3	26.357	0.803	29.292	0.825	29.058	0.820	29.280	0.819	24.180	0.848	27.560	0.811	29.443	0.864
4	17.149	0.755	17.260	0.770	17.549	0.766	24.913	0.798	26.185	0.811	26.616	0.735	26.817	0.840
5	16.590	0.813	17.897	0.816	18.405	0.823	25.028	0.824	25.236	0.845	24.847	0.812	26.417	0.838
6	15.386	0.786	16.006	0.800	16.693	0.810	24.801	0.814	25.940	0.843	25.450	0.802	26.081	0.834
7	15.769	0.699	17.337	0.708	16.967	0.703	24.388	0.686	25.540	0.536	24.992	0.618	25.989	0.783
8	15.842	0.711	16.920	0.731	16.993	0.715	24.166	0.727	25.167	0.778	24.710	0.738	25.677	0.782
9	14.726	0.702	14.248	0.727	14.462	0.704	22.938	0.723	24.483	0.774	24.813	0.733	25.212	0.780
10	16.908	0.697	17.823	0.708	19.272	0.729	25.319	0.738	25.025	0.751	25.567	0.748	24.988	0.767
Avg	18.111	0.777	20.476	0.795	21.302	0.793	27.331	0.795	26.724	0.802	27.407	0.782	28.158	0.835

on a GAN framework with an optimized Transformer-based generator significantly enhances image clarity and contrast, recovering high-frequency textures and background details. The integration of residual connections at each Transformer layer and the shifted window mechanism further strengthens long-range dependency modeling, ensuring more effective deep feature extraction. Based on this, our model can maintain both the foreground and background of feature areas without introducing artifacts, demonstrating exceptional robustness and practicality for high-precision NDT applications.

Tables I demonstrates the quantitive evaluation of SOTA models based on PSNR and SSIM metrics, highlighting our model's superior performance. Achieving the highest average SSIM of 0.835, our model surpasses Real-ESRGAN and SwinNet in image sharpness assessment. Furthermore, with an average PSNR of 28.158, it excels in maintaining structural integrity, particularly in complex regions. This improvement arises from integrating GAN with a transformer-based generator, combining GAN's ability to synthesize detailed features with the transformer's capability to capture global dependencies, surpassing traditional CNN-based GAN models.

C. Model Deployment

In Fig. 7, a web-based application was designed using HTML, CSS, and JavaScript, with a Flask API backend to manage model inference requests. The API allows users to set resolution enhancement factors and upload images for processing. Once an input image is received, the model enhances its resolution, encodes the result in Base64 format, and transmits it back to the website for display. This setup provides an efficient and user-friendly interface for resolution enhancement on the web, improving the processing time. For example, when scanning a power MOSFET sample with dimensions of 37 x 35 mm using the step size of 100 μm (resulting in an image resolution of 370 x 350 pixels), the scanning process takes approximately 2.42 minutes. Reducing the step size to 50 μm (which gives an image resolution of 740 x 700 pixels) would typically double the scanning time to about 4.84 minutes. However, by applying our proposed SR method, the resolution enhancement is achieved in just 16.44 seconds, reducing the

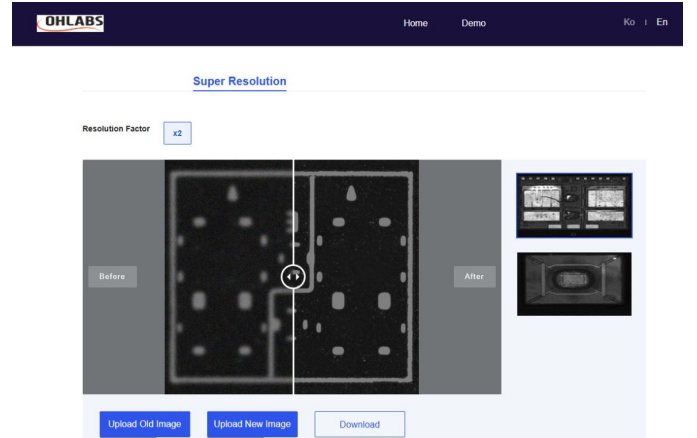


Fig. 7. Model Deployment on the web page.

processing time by 44.26%, while maintaining image quality equivalent to the 50 μm step size scan.

IV. CONCLUSION

In this study, we propose a GAN-based model with a transformer generator for SR in industrial ultrasound imaging, focused on enhancing image quality. Using a multi-stage training process that combines Pixel-wise, Perceptual, and Adversarial loss functions, our model achieves substantial gains in PSNR and SSIM metrics compared to traditional GAN models. The transfer learning captures general features, followed by fine-tuning specific SR tasks, resulting in high structural accuracy and enhanced perceptual quality. The addition of synthetic data through a degradation model, integrated with real LR images, further strengthens the dataset, enabling robust performance across varied imaging scenarios. Our future work will explore the development of a transformer-based version of the discriminator model, which will be tested and synchronized with the generator to evaluate its potential to improve model performance and accuracy. The training process will be designed to generate defect-oriented SR images, focusing on preserving the defect structure and morphology to further enhance industrial applicability.

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REFERENCES

- [1] Yutai Su, Jing Shi, Yuan-Ming Hsu, Dai-Yan Ji, Alexander David Suer, and Jay Lee. Volumetric nondestructive metrology for 3d semiconductor packaging: A review. *Measurement*, page 114065, 2023.
- [2] Hyunung Yu. Scanning acoustic microscopy for material evaluation. *Applied Microscopy*, 50(1):25, 2020.
- [3] Honggang Chen, Xiaohai He, Linbo Qing, Yuanyuan Wu, Chao Ren, Ray E Sheriff, and Ce Zhu. Real-world single image super-resolution: A brief review. *Information Fusion*, 79:124–145, 2022.
- [4] Sabina Umirzakova, Shabir Ahmad, Latif U Khan, and Taegkeun Whangbo. Medical image super-resolution for smart healthcare applications: A comprehensive survey. *Information Fusion*, 103:102075, 2024.
- [5] Pragyban Banerjee, Shivam Milind Akarte, Prakhar Kumar, Muhammad Shamsuzzaman, Ankit Butola, Krishna Agarwal, Dilip K Prasad, Frank Melandsø, and Anowarul Habib. High-resolution imaging in acoustic microscopy using deep learning. *Machine Learning: Science and Technology*, 5(1):015007, 2024.
- [6] Fei Gao, Bing Li, Lei Chen, Xiang Wei, Zhongyu Shang, and Chunman Liu. Ultrasound image super-resolution reconstruction based on semi-supervised cyclegan. *Ultrasonics*, 137:107177, 2024.
- [7] Zhihao Wang, Jian Chen, and Steven CH Hoi. Deep learning for image super-resolution: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 43(10):3365–3387, 2020.
- [8] Kyle Olszewski, Zimo Li, Chao Yang, Yi Zhou, Ronald Yu, Zeng Huang, Sitao Xiang, Shunsuke Saito, Pushmeet Kohli, and Hao Li. Realistic dynamic facial textures from a single image using gans. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 5429–5438, 2017.
- [9] Jianwei Yang, Chunyuan Li, Pengchuan Zhang, Xiyang Dai, Bin Xiao, Lu Yuan, and Jianfeng Gao. Focal attention for long-range interactions in vision transformers. *Advances in Neural Information Processing Systems*, 34:30008–30022, 2021.
- [10] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 1905–1914, October 2021.
- [11] Kai Zhang, Wangmeng Zuo, and Lei Zhang. Learning a single convolutional super-resolution network for multiple degradations. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3262–3271, 2018.
- [12] Kai Zhang, Jingyun Liang, Luc Van Gool, and Radu Timofte. Designing a practical degradation model for deep blind image super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4791–4800, 2021.
- [13] Kai Zhang, Wangmeng Zuo, and Lei Zhang. Deep plug-and-play super-resolution for arbitrary blur kernels. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1671–1681, 2019.
- [14] Chih-Yuan Yang, Chao Ma, and Ming-Hsuan Yang. Single-image super-resolution: A benchmark. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part IV 13*, pages 372–386. Springer, 2014.
- [15] Kai Zhang, Luc Van Gool, and Radu Timofte. Deep unfolding network for image super-resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3217–3226, 2020.
- [16] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 136–144, 2017.
- [17] Xintao Wang, Ke Yu, Chao Dong, and Chen Change Loy. Esrgan: Enhanced super-resolution generative adversarial networks. *arXiv preprint arXiv:1809.00219*, 2018.
- [18] Stéphane d’Ascoli, Hugo Touvron, Matthew Leavitt, Ari Morcos, Giulio Biroli, and Levent Sagun. Convit: Improving vision transformers with soft convolutional inductive biases. *arXiv preprint arXiv:2103.10697*, 2021.
- [19] Xiaoyang Zeng, Zizheng Pan, Shiji Song, Jiahui Mei, Yulin Yang, Shuyang Zhou, Zhiliang Wang, Hongsheng Li, Minghui Tan, and Gao Huang Wang. Cswin transformer: A general vision transformer backbone with cross-shaped windows. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12124–12134, 2022.
- [20] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10012–10022, 2021.
- [21] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.
- [22] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [23] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (gelu). *arXiv preprint arXiv:1606.08415*, 2016.
- [24] Wenzhe Shi, Jose Caballero, Ferenc Huszar, Johannes Totz, Andrew Aitken, Rob Bishop, Daniel Rueckert, and Zehan Wang. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1874–1883, 2016.
- [25] Edgar Schonfeld et al. A u-net based discriminator for gans. *NeurIPS*, 2020.
- [26] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks, 2018.
- [27] Edgar Schonfeld, Bernt Schiele, and Anna Khoreva. A u-net based discriminator for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8207–8216, 2020.
- [28] A. Pham et al. Advanced ndt imaging with the tsam-400 system. *Journal of Non-Destructive Testing*, 45(3):123–130, 2022.
- [29] Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, et al. Photo-realistic single image super-resolution using a generative adversarial network. *arXiv preprint arXiv:1609.04802*, 2017.
- [30] Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. In *European conference on computer vision*, pages 694–711. Springer, 2016.
- [31] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 126–135, 2017.
- [32] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1905–1914, 2021.
- [33] Yochai Blau and Tomer Michaeli. The perception-distortion tradeoff. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6228–6237, 2018.
- [34] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. Swinir: Image restoration using swin transformer. *arXiv preprint arXiv:2108.10257*, 2021.
- [35] Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch sgd: Training imagenet in 1 hour. *arXiv preprint arXiv:1706.02677*, 2017.
- [36] Xintao Wang, Liangbin Xie, Ke Yu, Kelvin C.K. Chan, Chen Change Loy, and Chao Dong. BasicSR: Open source image and video restoration toolbox. <https://github.com/XPixelGroup/BasicSR>, 2022.
- [37] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004.
- [38] Zhou Wang, Eero P Simoncelli, and Alan C Bovik. Modern advances in image quality assessment. *IEEE signal processing magazine*, 26(2):98–117, 2006.
- [39] Gabriel Gouvine. Super-resolution networks for pytorch. <https://github.com/Coloquinte/torchSR>, 2021.
- [40] Gabriel Gouvine. Ninasr: Efficient small and large convnets for super-resolution. <https://github.com/Coloquinte/torchSR/blob/main/doc/NinaSR.md>, 2021.
- [41] Byungjun Ahn, Hyun Cho, and Chung-Na Kim. Fast and accurate image super-resolution with deep residual networks. *IEEE Transactions on Image Processing*, 27(7):3224–3235, 2018.